

Discrete Mathematics

CSE 4203

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September 4, 2025

Outline

- 1 Definition of Trees
- 2 Rooted Trees
- 3 Properties of Trees
- 4 Applications

What is a Tree?

A **tree** is a connected, undirected graph with no simple circuits.

Equivalent definitions:

- A connected graph with no cycles.
- An acyclic graph that is connected.
- A graph with n vertices and $n - 1$ edges that is connected.

Examples

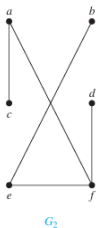
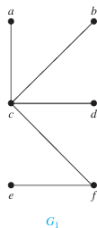
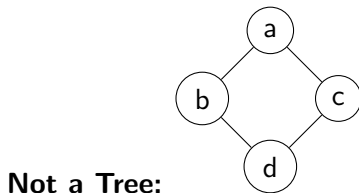
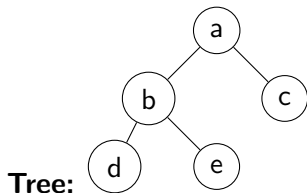


Figure: Determine the graphs that are tree or not trees.

Rooted Trees

A **rooted tree** is a tree with one designated vertex called the *root*.

- Every edge is directed away from the root.
- Concepts: parent, child, siblings, leaves, internal vertices.

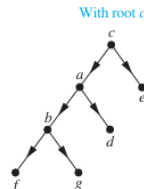
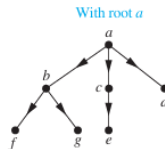
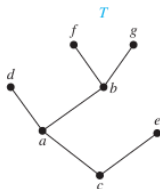
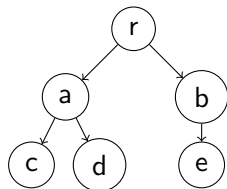


Figure: A tree and rooted trees formed by designating two different roots.

Origins of Terminology

- The terminology for trees has **botanical** and **genealogical** origins.
- We consider a **rooted tree** T .

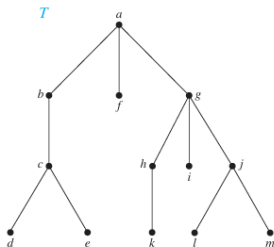


Figure: Rooted tree T

Tree Terminology

Basic Relations

- **Parent:** For a vertex v (not root), its parent is the unique vertex u with an edge $u \rightarrow v$.
- **Child:** If u is parent of v , then v is a child of u .
- **Siblings:** Vertices sharing the same parent.

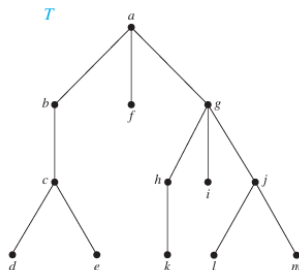


Figure: Rooted tree T

Hierarchy

- **Ancestors:** Vertices on path from root to v (excluding v).
- **Descendants:** Vertices that have v as ancestor.

Tree Terminology

Vertex Types

- **Leaf:** A vertex with no children.
- **Internal vertex:** A vertex with children.
- **Root:**
 - Internal if tree has more than one vertex.
 - A leaf if it is the only vertex.

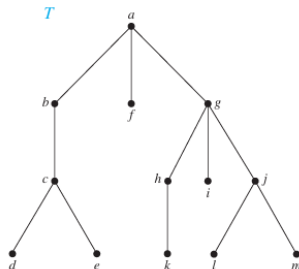


Figure: Rooted tree T

m -ary Trees: Definition and Example

Definition 3

A rooted tree is called an **m -ary tree** if every internal vertex has no more than m children.

The tree is called a **full m -ary tree** if every internal vertex has exactly m children.

An m -ary tree with $m = 2$ is called a **binary tree**.

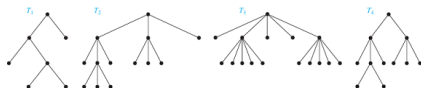


Figure: Four rooted trees.

Solution

- T_1 is a **full binary tree**: each internal vertex has 2 children.
- T_2 is a **full 3-ary tree**: each internal vertex has 3 children.
- T_3 is a **full 5-ary tree**: each internal vertex has 5 children.
- T_4 is **not** a full m -ary tree: some internal vertices have 2 children and others 3.

Ordered Binary Tree

Definition

In an **ordered binary tree** (usually called a **binary tree**):

- If an internal vertex has two children, the first child is called the **left child** and the second child is called the **right child**.
- The tree rooted at the left child of a vertex is called the **left subtree**.
- The tree rooted at the right child of a vertex is called the **right subtree**.

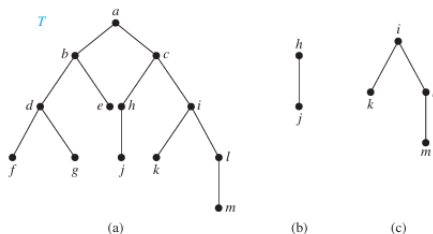


Figure: A binary tree T and left and right subtrees of the vertex c .

Level and Height of a Rooted Tree

Definitions

- The **level** of a vertex v in a rooted tree is the length of the unique path from the root to v .
- The **level of the root** is defined as 0.
- The **height** of a rooted tree is the maximum of the levels of its vertices.
- Equivalently, the height is the length of the longest path from the root to any vertex.

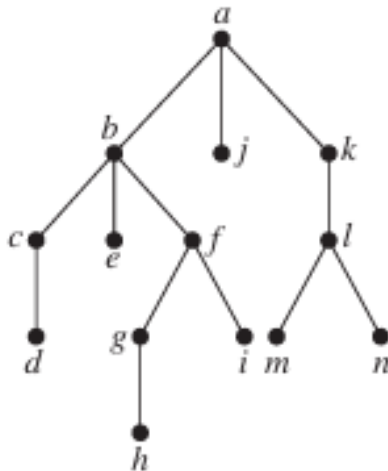
Example 10: Level and Height of a Rooted Tree

Problem

Find the level of each vertex in the rooted tree shown in Figure 13. What is the height of this tree?

Solution

- Root a : level 0
- Vertices b, j, k : level 1
- Vertices c, e, f, l : level 2
- Vertices d, g, i, m, n : level 3
- Vertex h : level 4
- **Height of the tree: 4**



Key Properties

If T is a tree with n vertices ($n > 1$):

- T has exactly $n - 1$ edges.
- There is a unique path between any two vertices.
- Adding an edge creates exactly one cycle.
- Removing any edge disconnects the tree.

Number of Edges

Theorem: A tree with n vertices has exactly $n - 1$ edges.

Proof by induction on n .

Base case. For $n = 1$, the tree has a single vertex and no edges, so $0 = 1 - 1$.

Inductive step. Assume every tree with $n - 1$ vertices has exactly $(n - 1) - 1 = n - 2$ edges. Let T be a tree on n vertices. Every finite tree has a leaf (a vertex of degree 1). Remove a leaf v and its incident edge e to obtain T' . Then T' is still a tree: it is connected (removing a leaf cannot disconnect the remaining vertices) and acyclic (removing edges cannot create cycles). T' has $n - 1$ vertices, so by the inductive hypothesis it has $n - 2$ edges. Adding back v and e yields T with $(n - 2) + 1 = n - 1$ edges. □

Full m -ary Tree

Theorem 3

A full m -ary tree with i internal vertices contains

$$n = mi + 1 \text{ vertices.}$$

Example (Optional)

For $m = 2$ (binary tree) and $i = 3$ internal vertices, the tree has

$$n = 2 \cdot 3 + 1 = 7 \text{ vertices.}$$

Proof of Theorem 3 (Induction)

Proof (by induction on i)

Base case: $i = 1$

- One internal vertex has m children.
- Total vertices $n = 1 + m = m \cdot 1 + 1$.

Inductive step: Assume the formula holds for $i = k$, so $n = mk + 1$.

- Consider a full m -ary tree with $k + 1$ internal vertices.
- Remove one internal vertex along with its m children $\rightarrow$ remaining tree has k internal vertices.
- By the inductive hypothesis, remaining tree has $mk + 1$ vertices.
- Adding back the removed vertex and its m children gives $(mk + 1) + m = m(k + 1) + 1$ vertices.

Conclusion: By induction, $n = mi + 1$ holds for all $i \geq 1$.

Theorem 4: Full m -ary Tree Relationships

Theorem 4

Let a full m -ary tree have n vertices, i internal vertices, and l leaves. Then:

(i) Given n vertices:

$$i = \frac{n-1}{m}, \quad l = \frac{(m-1)n+1}{m}$$

(ii) Given i internal vertices:

$$n = mi + 1, \quad l = (m-1)i + 1$$

(iii) Given l leaves:

$$n = \frac{ml-1}{m-1}, \quad i = \frac{l-1}{m-1}$$

Derivation of Theorem 4

Derivation

Let a full m -ary tree have n vertices, i internal vertices, and l leaves.

- In a full m -ary tree, each internal vertex has m children.
- Total edges = total children = mi
- Also, in any tree, edges = $n - 1$

$$n - 1 = mi \implies i = \frac{n - 1}{m} \quad (\text{part i})$$

- Number of leaves: $l = n - i = n - \frac{n-1}{m} = \frac{(m-1)n+1}{m}$ (part i)
- Given i internal vertices: $n = mi + 1$,
 $l = n - i = (mi + 1) - i = (m - 1)i + 1$ (part ii)
- Given l leaves: $i = n - l$,
 $n = i + l = (m - 1)i + 1 + i = mi + 1 \implies n = \frac{ml-1}{m-1}, i = \frac{l-1}{m-1}$ (part iii)

Chain Letter Problem

Problem

Suppose someone starts a chain letter. Each person who receives the letter is asked to send it to four other people.

Some people send it, while others do not send any letters.

- ① How many people have seen the letter, including the first person, if no one receives more than one letter and the chain ends after 100 people read it but did not send it out?
- ② How many people sent out the letter?

Solution Outline

- Model the situation as a **full 4-ary tree**, where:
 - Internal vertices = people who sent letters
 - Leaves = people who did not send letters
- Given: Number of leaves $l = 100$, $m = 4$
- Recall formula for a full m -ary tree:

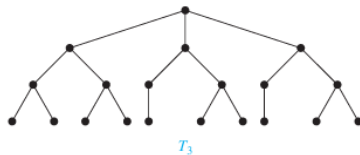
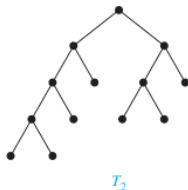
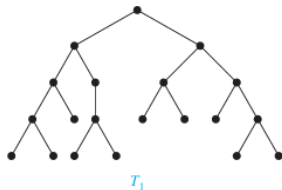
Balanced m -ary Tree Example

Definition

A rooted m -ary tree of height h is **balanced** if all leaves are at levels h or $h - 1$.

Solution

- T_1 is balanced: all leaves at levels 3 and 4.
- T_2 is not balanced: leaves at levels 2, 3, and 4.
- T_3 is balanced: all leaves at level 3.



Theorem on Leaves in an m -ary Tree

Theorem 5.

In any rooted m -ary tree of height h , there are at most m^h leaves.

Proof (Sketch).

- **Base case ($h = 0$):** A tree of height 0 has only the root, so at most $1 = m^0$ leaf.
- **Inductive step:**
 - Assume true for height $h - 1$.
 - A tree of height h : the root has at most m children.
 - Each child's subtree has at most m^{h-1} leaves.
 - So total leaves $\leq m \cdot m^{h-1} = m^h$.

Example: In a full binary tree ($m = 2$) of height $h = 3$, maximum leaves $= 2^3 = 8$.

Minimum Number of Leaves in a Full m -ary Tree

Theorem: A full m -ary tree of height h has at least

$$L_{\min} = h(m - 1) + 1 \text{ leaves.}$$

Reasoning:

- In a **full m -ary tree**, every internal node has exactly m children.
- To minimize leaves, make the tree a **spine**: one path of length h .
- At each internal node along the spine:
 - Choose **1 child** to continue the path.
 - The other $(m - 1)$ children become **leaves**.
- There are h internal nodes along the spine, each contributing $(m - 1)$ leaves.
- The final node at depth h is also a leaf.

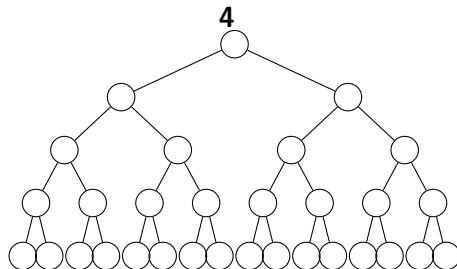
$$L_{\min} = h(m - 1) + 1$$

Complete m -ary Trees

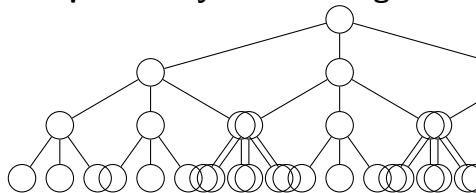
Definition. A *complete m -ary tree* is a full m -ary tree in which every leaf is at the same level.

Examples:

Complete Binary Tree of Height



Complete 3-ary Tree of Height 3

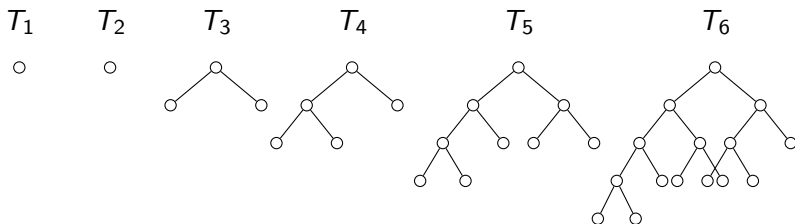


Rooted Fibonacci Trees

Definition. The rooted Fibonacci trees $\{T_n\}$ are defined recursively:

- T_1 and T_2 are single-vertex trees.
- For $n \geq 3$, T_n is a root whose
 - left subtree is T_{n-1} ,
 - right subtree is T_{n-2} .

The first six Fibonacci trees:



Applications of Trees

- Representing hierarchical data (organization charts, file systems).
- Decision trees in algorithms.
- Network routing.
- Expression trees in compilers.
- Spanning trees in graph theory.

Summary

- Trees are connected, acyclic graphs.
- Unique path between any two vertices.
- Rooted trees introduce hierarchy.
- Widely used in computer science and networks.